P510/1
PHYSICS
Paper 1
(Theory)
Jul/Aug, 2026
3 hours

ALLIANCE JOINT EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

PHYSICS

(Theory)

Paper 1

3 hours

INSTRUCTIONS TO CANDIDATES:

*This examination paper consists of two sections; **A** and **B**. It has **six** items.*

*Section **A** has **two compulsory** items.*

*Section **B** has two parts; **I** and **II**. Respond to only **one** item from **each** part.*

*Respond to **four** items in all.*

*Any additional item(s) responded to, will **not** be scored.*

*Responses to **each** item should start on a **fresh page**.*

Silent non-programmable scientific calculators may be used.

TURN OVER

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SECTION A

Respond to all items in this section

ITEM 1.

Uganda has made significant strides in space technology marked by the successful launch of its first Satellite-Pearl AfricaSat-1 on November 7, 2022. The project manager is intensifying mechanisms to develop more infrastructure to enable a second satellite launch before 2028. At Mpoma East station- Mukono, a satellite of mass 200kg is to be launched in a parking orbit and Project Manager needed to establish the height of the parking orbit, the mechanical energy of this satellite and get understanding the effect of reducing the mechanical energy of the satellite on its safety.

This information was necessary to establish safety precautions of the satellite in the orbit. The satellite is considered safe if its mechanical energy does not exceed **-9.0×10^{9}J**. The project manager called one of the space Engineers in charge who had gone back to his nearby home to rush and return not later than 5 minutes to do the necessary calculations and deliver information. The engineer had parked his car on a road which was a straight stretch up to the station. He wants to start the engine of the car and give it a power of 22kW for a driving force of 1.1kN to be developed so as to uniformly accelerate for the first 1.2km just before applying brakes for the remaining 0.5km to decelerate uniformly to rest at the station.

The project supervisor also had tasked the engineer to explain how he can utilize some locally made materials to design a device that can produce a lift force with a mounted camera to capture aerial photos at the station. As the engineer had may other pressing issues at hand to solve.

Hint:

Coefficient of friction on the used road is 0.025
Mass of the car = 2 tones
Acceleration due to gravity $g = 9.81ms^{-2}$
Radius of the Earth R, $E = 6.4\times10^{6}m$

Task:

You have been selected to help;
(a) Analyze the data to provide the necessary information needed by the project manager and the establish if the engineer can reach in time to deliver it.
(b) Explain how the device needed by the supervisor can be made and able to serve the purpose.

ITEM 2

Cancer in Uganda is a major, increasing public health crisis. One of the Privately owned Hospital wants to start a cancer treatment department. The proposed cost effective method is the use of an efficient Cobalt ($^{60}_{27}Co$) teletherapy machine which utilizes gamma rays from radioactive cobalt decay to destroy tumors. The radioactive isotope

$^{60}_{27}$**Co** decays to Nickel **(Ni)** producing a beta particle and the gamma rays. The project manager sat with some health workers who suggested that before the machine is bought, there should be a clear understanding of the stability of atoms using binding energy per nucleon graph, an equation showing the decay, a determined rate of disintegration and the amount of energy the machine can possibly produce un the process.

The available reference data indicate that an efficient teletherapy machine exploits 20g of the Cobalt at a rate of disintegration not more than 9×10^{14} Bq to produce at least **5.5×10^{20} MeV** energy. They told the manager to involve a student of Senior Six advanced level physics for help if the best solutions were to be obtained.

The project manager was told that the student should help in describing the structure and operation of a device which uses particles of same properties like the beta particle to produce radiations for use in scanning whether cancer patients who always complain of painful bones have fractures.

Hint:

Half-life of $^{60}_{27}$Co is 5.27 years
Mass of $^{60}_{27}$Co = 59.93381U
Mass of Ni = 59.93079U
Mass of an electron = 0.0005U where 1U = 931MeV

Task:

You have been selected for help

(a) analyze the given data to address the concerns raised by health workers

(b) Explain how the device suggested by the manager can work to serve the desired purpose

SECTION B

Part I

Attempt **one** *item from this part*

ITEM 3.

Students in a certain secondary school complained of too much heat in the classroom. They claimed that this could be due to much light which entered through the door which would remain open when one entered or the quality of the newly installed windows. The head teacher convened a meeting with the class and invited some S.6 physics students, their teacher of physics and the supplier of the windows to address the issues raised. The supplier said that the windows supplied are of good quality and each window had a double-glazed glass consisting of two Sheets of glass 5mm thick separated by a layer of air 10mm thick.

The cross-sectional area of the window was **$2m^2$** and absorbed 5% of radiation from the sun of solar constant **$1100Wm^{-2}$.** He also said that the thermal conductivity of the used materials were **$0.025Wm^{-1}K^{-1}$** for air and **$0.8Wm^{-1}K^{-1}$**. He said that all this ensures that even when the outside surface of the window was at 35°C, the inside temperature in the classroom would be maintained at most 20°C which is good enough for the conducive atmosphere.

The physics teacher raised doubt on whether said inside temperature in the classroom and the thermal conductivity of the glass are accurate. The teacher requested one of the physics students to explain how an experiment can be done using a piece of glass material used to confirm the value and also establish whether the inner class room temperature was at most **20°C**.

The teacher also requested the same student to explain how the outer surface of the window can be made in order to reduce the amount of energy absorbed from light falling on it. The head teacher remembered that there is a simple device which when installed on the door, it enables it to slowly close automatically after it is opened.

Task:
You have been selected by the physics teacher,
(a) Analyze the data to address the requests put forward to calm down the class members about window quality.
(b) Assess the information given to explain is mode of operation of the said device.

ITEM 4.
Musical instruments such as guitars are widely used in Uganda for music to serve as instrument of performance, education and entertainment. Ojok is a famous business man selling Musical instruments in town. Of recent past, his customers have been complaining of the effect of heating in strings of the guitars. Ojok sent his S.6 physics child who had come to help him on the weekend at the shop to call the Luthier (guitar repairer) to check if customers claims were right. The luthier tested a fixed string of length **40cm** and mass **$1.2x110^{-3}kg$** on one of the guitars. He tuned the string to produce a third harmonic of Frequency **750Hz**. The Luthier requested the S.6 candidate to determine tension which was developed in the string to address customer concerns and explain the effect in the string experienced.

Luthier also tasked the candidate to derive the equation of the wave generated in fixed strings relating to amplitude and wave displacement such that future string properties can be studied. As the Luthier was busy fixing the guitar problems, Ojok moved at the

Balcony viewing the day moon using a refracting device in normal adjustments. The device was a gift to him and he was excited at his observation. Meanwhile Ojok was taking a cold soda in a glass bottle and by the time he picked the bottle for another sip of a soda, some liquid had collected outside the bottle which kept him wondering whether the glass had pores. He immediately called his S.6 candidate to provide explanations.

Hint: *The guitar string breaks if its tension exceeds 100N.*

Task:
You are the selected student;

- **(a)** Help the luthier to address customer concerns and obtain the desired expression for future use and explain what happened on the outside the soda bottle
- **(b)** Describe to Mr Ojok the structure and operation of the device he received as a gift.

Part II

Attempt **one** *item from this part*

ITEM 5.

Some schools in Uganda are faced with challenges of ineffective communication especially on assemblies due to lack of address systems. The head teacher in one of the Secondary Schools bought a portable speaker address system. This speaker could however, work for a short time and stop. The head teacher claimed that the problem of the speaker could be a high internal resistance of the battery used for charging.

Some learners claimed that it was the charging time which was short. The recommended charging time for the power bank to be full should be at least 15 minutes. One S.5 student approached the head teacher and requested to test the battery and establish if it was indeed the problem. The student connected the battery in series with a variable resistor and an ammeter. When the resistor was set 21Ω, the current through it was 0.48A and when it was set to 36Ω the current was 0.30A. The power bank in the speaker was made up of two super capacitors each of capacitance 1.0F in series. The head teacher further intervened and asked how the capacitors got charged by the battery and the time one needed to wait before the speaker is fully charged.

The Deputy dead teacher on the other hand asked the student to explain how he could change the a.c from other sources into d.c for use in powering some other school gadgets. The S.5 student tried but could not succeed in addressing any of the matters raised. He went to S.6 physics students for assistance.

Hint: *A good battery for charging has internal resistance not more than 20Ω.*

Task:

You have been selected to help:

(a) analyze the information given in the scenario and address the claims and concerns of the head teacher.

(b) Address the response of the deputy head teacher.

ITEM 6.

Listening to religious radios in highly advisable for older adults to promote emotional comfort and mental stimulation. Opio visited his parent with his child who was doing physics in S.6. The parent complained to Opio of the challenge they had with their radio set which often failed to tune in their favorable religious program. The radio operated at a frequency of 61.2MHz. Opio opened the radio and found its tuning circuit as indicated in ***figure 6(i)*** very weak.

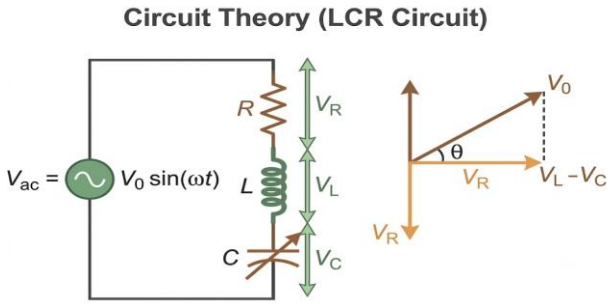


Figure 6 (i): Resonance circuit for tuning radio.

Opio decided to install a new stronger variable capacitor, C and a pure inductor, L. He purchased materials and made the two smaller capacitors each of plate separation 2cm and maximum area of overlap 5.0×10^{-4} m² filled with a material of dielectric constant 4. The two capacitors were joined in series and connected with a rotor for tuning.

He made L from a 1.0cm air-cored solenoid of 10 turns and 1.2×10^{-3} m² cross sectional area. Opio requested his child to explain the effect of a dielectric in the capacitors, determine the magnetic field density developed by L and perform necessary calculations to establish whether the new circuit can tune to the desired religious channel

Opio's Parent also took advantage of the presence of the grand child and requested to make them understand the structure and operation of a device used by used in charging the radio because, this parent had always wondered how the charger can be connected in the socket which supplies 240V to charge a radio which operates at only 12V.

Hint:

Permeability in free space $\mu_0 = 4\pi \times 10^{-7}\ H\ m^{-1}$

Permittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12}\ F\ m^{-1}$

Task: As a S.6 physics student:

(a) analyze the information in the scenario to answer the requests made by Opio

(b) Provide the solution made by Opio's parent.

END